Code No: 5502

OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)

.E. (CSE, AI&ML) III - Semester (MAIN) Examinations Dec-2024/Jan 2025

DATA STRUCTURES

: 3 hours

Max. Marks : 60

: *i) Answer* ***Question No. 1*** *is compulsory and answer any* ***four questions*** *from the remaining questions (2- 7), each Question carries 12 Marks.*

ii) Answers must be written in same order as they occur in the Question Paper.

iii) ***Missing data, if any, may suitably be assumed.***

		Marks	BT	CO
a)	Explain the difference between polynomial time and exponential time with an example.	2	2	2
b)	Discuss the differences between arrays and linked lists. Mention one scenario where linked lists are preferred.	2	2	2
c)	Write the adjacency matrix representation for a graph with 4 nodes and edges: A-B, A-C, B-D, C-D.	2	3	1
d)	Describe the purpose of balancing in AVL trees. Why is it necessary.	2	3	3
e)	Differentiate between Depth-First Search (DFS) and Breadth-First Search (BFS) in graphs.	2	4	2
f)	Derive the recurrence relation for Merge Sort and solve it using the Master Theorem.	2	4	3
2. a)	Solve the following recurrence relation using iteration and then verify the solution using Master Theorem: $T(n) = 4T\left(\frac{n}{2}\right) + n^2$	6	4	1
b)	Demonstrate the difference between row-major and column-major storage for a 3x3 matrix of Lower Triangular Matrix.	6	3	1
3. a)	Write the algorithm for converting an infix expression to a postfix expression. Convert (A + B) * (C / D) using the algorithm.	6	3	2
b)	Explain how a linear queue differs from a circular queue. Provide an example of its application.	6	2	2
4. a)	Construct a binary tree for the following expression: ((A + B) * C) - D. Write its pre-order and in-order traversals.	6	3	3

	b) Define and explain the importance of B trees. Illustrate its structure with a small example data of 10 elements to generate 2-3BTree	6	4	3
5.	a) Demonstrate the working of Kruskal's algorithm to find the MST of a graph with the following edges and weights: $A-B:3, A-C:1, B-C:2, B-D:5, C-D:4$	6	3	4
	b) Define a Minimum Spanning Tree (MST). Why can't a graph with negative cycles have an MST.	6	5	4
6.	a) Demonstrate the binary search algorithm with a step-by-step example for the array: [3, 6, 9, 12, 15, 18, 21]. Search for the number 15 by writing an algorithm	6	3	2
	b) Compare Merge Sort and Heap Sort. Which one is more suitable for external sorting and why.	6	4	3
7.	a) Define hashing. Construct a hash table for the following keys using modulo-5 hashing and linear probing: 12, 25, 36, 20, 8.	6	3	4
	b) Explain BFS algorithm. Explain traversal by tracing with an example.	6	3	3

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